BGIN Block#11 Meeting Report

- BGIN is a multi-stakeholder discussion body / distributed think tank.

Layer2 Governance

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BGIN Block#11

Layer2 Governance

Date: Monday, October 21, 2024

Location: Washington DC, United States

Duration: Approx. 90 minutes

Summary

Introduction and Setup

- **Starting and Setup Issues:** The presentation is set to begin after some initial setup delays. Discussions on screen-sharing and technical issues are mentioned.
- **Casual Session Setup:** Small audience size, allowing for a more casual and interactive presentation style.

Layer 2 Governance and Definitions

- **Definition Ambiguity of L2:** The definition of Layer 2 (L2) solutions is ambiguous. Historically, L2 was viewed as a scaling solution for Layer 1 (L1) challenges like high fees and slow transaction speeds by handling transactions off the L1 chain.
- **Challenges with L2 Definition:** L2 solutions were intended to reduce the volume of L1 transactions by only recording final states. The definition of "off-chain" is also ambiguous and needs to be clarified as “off L1 chain.”
- **Potential Marketing Misuse:** "L2" has become a marketing term. Many projects call themselves L2 if they provide faster transactions and lower fees than L1, often without meeting core criteria.

L2 Characteristics and Scaling

- **Traditional L2 Characteristics:** L2s are expected to facilitate transactions faster and cheaper than L1, support the same virtual machine as L1 for application compatibility, and store a summarized version of L1 transactions.
- **Scaling and On/Off Ramp Transactions:** L2 solutions allow for direct on- and off-ramp transactions (fiat and crypto exchanges), bypassing L1. This raises questions about whether these are truly L1 scaling solutions.

- **L2 Development Freedom:** Engineers can develop applications on L2 using the same language and tools as L1, allowing for smooth interoperability.

Governance in L2

- **Independent Governance:** L2s often have their own governance, sometimes through decentralized autonomous organizations (DAOs), using native or governance tokens. This structure is similar to L1 governance.
- **Centralization Concerns:** There is ongoing debate if such self-governed L2s are simply a form of L1. Their self-identification as L2 may be a strategy to imply they offer L1-level security.

Understanding Sequencers and Data Storage in L2

- **Sequencers in L2:** Sequencers are responsible for validating transactions on L2 and submitting proof to L1. This role is essential for maintaining integrity between L2 and L1.
- **Data Storage and Cost Challenges:** Storing all L2 transaction data on L1 is costly due to limited L1 capacity. To address this, some L2 solutions use off-chain data storage layers for data availability without burdening L1 capacity.
- **Regulatory Touchpoints:** Sequencing and data storage are critical regulatory touchpoints. If L2 transaction data is hidden, users lose the ability to validate transactions, leading to potential issues with integrity and compliance.

Key Concepts and Processes in Sequencing

1. **Entering Phase:** Users lock assets in L1, and sequencers enable equivalent assets on L2 for use.
2. **Usage Phase:** Sequencers create a state root of L2 transactions and submit it to a roll-up contract on L1, frequently updating the state.
3. **Exiting Phase:** When users want to withdraw, they create a Merkle proof to confirm their transaction history on L2 before unlocking assets on L1.

Proof Mechanisms: Fraud Proof vs. Validity Proof

- **Fraud Proof:**
  - Users submit a Merkle proof, and challengers can submit a fraud proof if they detect discrepancies.
  - Fraud proof requires a challenge period where finality is delayed to allow challenges.
- **Validity Proof:**
  - Sequencers create zk-proofs that validate L2 transactions on L1, removing the need for user-submitted fraud proofs.
  - This method is more efficient and provides continuous validation.

Risks in L2 Sequencing and Governance

- **Smart Contract Vulnerabilities:** Smart contracts used for sequencing could fail or be hacked, posing risks for both fraud and validity proofs.
- **Centralization of Sequencers:** A single sequencer creates a central point of failure. Efforts in the Ethereum community aim to increase the number of sequencers for decentralized governance.
- **Challenge Period and Finality Risk:** In fraud-proof systems, users must wait for transaction finality until the challenge period ends. If sequencers misbehave and no one notices, users risk losing assets.
- **Deposit Incentive Risks:** Users who submit false transactions can lose their deposits if successfully challenged. A well-incentivized system is needed to discourage malicious behavior.

Data Storage and Efficiency

- **Call Data and EIP-4844**:
  - L2 transaction data stored in L1 using "call data" is more efficient than other methods but still costly.
  - EIP-4844 introduces "blob data" for temporary, efficient storage, allowing better scalability for L2 solutions.
- **Data Availability Layers and Governance**:
  - These layers store L2 transaction data off L1, but must be monitored to avoid data tampering.
  - Governance tokens are sometimes used to incentivize accurate data monitoring, with penalties for misbehavior.

Risks and Regulatory Considerations

- **Smart Contracts and Sequencer Centralization**:
  - Both sequencer failure and smart contract vulnerabilities remain critical risk factors.
  - Data availability layers and governance mechanisms help mitigate some risks but introduce centralization concerns.
- **Fraud Proofs vs. Validity Proofs**:
  - Fraud proofs rely on challenge periods, where users can challenge invalid transactions.
  - Validity proofs, often utilizing zk-proofs, provide immediate validation but require high computational resources.

Staking and Validator Risks

- **Delegation Risks**:
  - Validators receiving delegated assets face unique regulatory and risk concerns. If slashing occurs, delegated assets might be lost, raising the need for regulatory oversight.
  - Pure staking with one's own tokens differs from delegated staking, as the latter involves handling others' assets and potentially greater regulatory touchpoints.

Money Laundering Risks in Layer 2 and Lightning Networks

- **Money Laundering through L2s**:
  - Repeated entry and exit from L2s could obscure transaction origins, complicating monitoring efforts.
  - Lightning Network transactions are harder to trace due to privacy enhancements, presenting unique challenges for law enforcement.

Optimistic Rollups and Security

- **Challenge System in Optimistic Rollups**:
  - Users can submit Merkle proofs as challenges if they suspect a sequencer has submitted invalid data.
  - Challenges are verified via smart contracts, and penalties apply for invalid challenges, promoting secure and transparent operations.

Open Questions and Final Thoughts

- **Centralized Sequencers and Finality**:
    - L2 solutions might still rely on L1 for finality, affecting their independence as truly decentralized solutions.
    - There's discussion on how to balance efficient processing with decentralized validation and security.
- **Engagement with the Ethereum Community**:
    - Future discussions will explore solutions for the identified risks, particularly with ongoing developments in Ethereum's community, such as proposer-builder separation.

Next Steps

- **Gathering Community Feedback**:
    - A proposal to solicit public contributions, particularly from the Ethereum community, aims to refine risk analyses and possible solutions.